EP8199 - GLOBAL OPTIMIZATION OF MIXED-INTEGER AND NON-CONVEX PROGRAMS
COLLOQUIUM 2

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Outline

- RECAP:
- MIXED-INTEGER LINEAR PROGRAMMING 1 - PROBLEM FORMULATION
- THE PRACTICAL VALUE OF MILPs
- STRENGTH OF MILP FORMULATION
- BRANCH-AND-BOUND ALGORITHM

RECAP I

1. Convex Sets: Line segment joining any two points contained inside set
2. Convex Functions: Convex Set. Function always under secant.
3. Convex Programs: Feasible set is a convex set. Objective function is a convex function.

Consider the NLP:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{h}(\mathbf{x}) = \mathbf{0} \\ & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x} \end{aligned} \tag{NLP-EQ-INEQ}$$

- The NLP is convex if f and g are convex functions and h is an affine function i.e., nonlinear equality constraints are non-convex.

RECAP II

- Another way to reason about this:

Theorem (Convex Sublevel Sets)

Let $C \subset \mathbb{R}^n$ be a nonempty convex set and let $g : C \to \mathbb{R}$ be a convex function. Then $L_\alpha^- = \{\mathbf{x} \in C : g(\mathbf{x}) \leq \alpha\}$ is a convex set for any $\alpha \in \mathbb{R}$, i.e., the sublevel sets of a convex function are convex sets.

VISUAL:

$$\begin{aligned} \Delta\dot{x} &= A\Delta x + B\Delta u \\ \Delta y &= C\Delta x + D\Delta u \end{aligned} \tag{1}$$

Cost function: $z^T Q_1 z + u^T Q_2 u$

RECAP III

CLASSIFICATION OF OPTIMIZATION PROBLEMS

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & f(\mathbf{x},\mathbf{y}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x},\mathbf{y}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x}, \mathbf{y} \in Y \equiv \{0,1\}^{n_y} \end{aligned} \tag{GP}$$

where:

- $\mathbf{x}$ and $\mathbf{y}$ denote the continuous and binary decision variables respectively
- the set $U \equiv [0,1]^{n_y}$ is the convex hull of the set Y (i.e., the smallest convex set that contains every point in Y
- $f : X \times U \to \mathbb{R}$ is the continuous objective function, and $\mathbf{g} : X \times U \to \mathbb{R}^m$ corresponds to continuous functions defining the constraints

RECAP IV

- **Linear Programming (LP)** - if there are no binary decision variables $\mathbf{y}$ and the functions f and $\mathbf{g}$ are linear
- **Mixed-Integer Linear Programming (MILP)** - if the functions f and $\mathbf{g}$ are linear.
- **Convex Nonlinear Programming** - if there are no binary decision variables $\mathbf{y}$, f and $\mathbf{g}$ are convex functions.
- **Mixed-Integer Convex Programming (MICP)** - if the functions f and $\mathbf{g}$ are convex functions
- **Nonconvex Nonlinear Programming (NLP)** - if there are no binary decision variables $\mathbf{y}$ and atleast one of the participating functions is nonconvex.
- **Nonconvex Mixed-Integer Nonlinear Programming (MINLP)** - if atleast one of the participating functions is nonconvex.

RECAP V

ALGORITHMS AND RESOURCES:

- **LP**: Simplex, Interior-Point
- **MILP**: Cutting plane, Branch-and-Bound, Branch-and-cut, Decomposition (e.g., Benders Decomposition if special structure)
- **Convex Nonlinear Programming**. Newton variants, Interior Point, Sequential Quadratic Programming.
- **MICP** - Branch-and-Bound, Generalized Benders Decomposition (special structure), Outer-Approximation,
- **Nonconvex Nonlinear Programming (NLP)** - Spatial Branch-and-Bound.
- **Nonconvex Mixed-Integer Nonlinear Programming (MINLP)** - Branch-and-Bound, Non-convex Generalized Benders Decomposition, Non-convex Outer Approximation

RECAP VI

The general procedure is to determine a sequence of **Lower Bounds LBDs** and **Upper Bounds UBDs** that bracket the optimal value. Converge the best known UBD and LBD such that the gap **UBD - LBD** is within a tolerance. Solvers usually report this gap.

Upper Bounding Techniques:

1. Determine any **feasible** point. Value of objective provides an UBD.
2. **Restriction**: Shrink the feasible set of an optimization problem by constraining it further. The optimal value or value or objective function value at any feasible point yield an UBD for the original optimization problem.

Lower Bounding Techniques:

1. **Relaxation** of constraints
2. **Convexification**
3. Using **Duality theory**
4. **Interval Analysis**

RECAP VII

1. Relaxation of Constraints Enlarging the feasible set of an optimization problem by relaxing constraints yields a lower bound. This is termed the Lower Bounding Principle:

Theorem (Lower Bounding Principle)

Suppose we have sets D and $E \subset \mathbb{R}^n$ such that $D \subset E$. Then

$$\min\{f(\mathbf{x}) : \mathbf{x} \in E\} \leq \min\{f(\mathbf{x}) : \mathbf{x} \in D\}$$

MILP Problem Formulation I

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An MILP is written as:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & \mathbf{c}^{\mathrm{T}}\mathbf{x} + \mathbf{d}^{\mathrm{T}}\mathbf{y} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}_{+}^{n_x},\ \mathbf{y} \in \mathbb{Z}_{+}^{n_y} \end{aligned} \tag{MILP}$$

where $\mathbb{R}_+$ is the set of nonnegative real numbers, $\mathbb{Z}_+$ is the set of nonnegative integers, $\mathbf{c} \in \mathbb{R}^{n_x}$, $\mathbf{d} \in \mathbb{R}^{n_y}$, $\mathbf{A} \in \mathbb{R}^{m \times n_x}$, $\mathbf{B} \in \mathbb{R}^{m \times n_y}$ and $\mathbf{b} \in \mathbb{R}^m$

Integer variables can be converted to binary variables:

MODELING WITH MILPs I

1. Binary choices
2. Disjunctive constraints
3. Fixed-charge problem
4. Dependent decisions
5. Logical relationships

Classical Problems: Knapsack problem, Transportation problem, Transshipment problem, Assignment problem, Set covering, packing and partitioning problems, Facility location problem, Vehicle routing problem, Traveling salesman problem

Ref: CH. 10. Dimitris Bertsimas, John Tsitsiklis. Introduction to Linear Optimization.

MODELING WITH MILPs II

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1. BINARY CHOICES - THE 0-1 KNAPSACK PROBLEM

Problem: You managed to break into a treasure cave with a knapsack which can hold a weight W. There are n valuable items in the cave, each item indexed by j has a weight w_j and value c_j. Which items would you pick?

MODELING WITH MILPs III

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2. DISJUNCTIVE CONSTRAINTS

Problem: You are required to invest in one of two machines $M1$ or $M2$ which take in two raw materials A or B. $M1$ converts 90% of A and 80% of B while $M2$ converts 85% of A and 82% of B. The selling price of the product is 10 $. How would you formulate this into a **single** optimization problem?

- **Big-M Reformulation**: Introduce one binary variable y_j for each disjunctive constraint j such that $\sum_j y_j = 1$. Relax the constraints by adding/subtracting $M(1 - y_j)$, where M is an appropriate large number.

- **Convex hull reformulation**

MODELING WITH MILPs IV

3. FIXED CHARGE PROBLEM: Often, many investments have a fixed charge of installation with other costs varying with size x of equipment: e.g.,

$$Cost(x) = \begin{cases} 0 & \text{if } x = 0 \\ a + bx & \text{if } x = 0 \end{cases}$$

$$\begin{aligned} \min_{x,y} \quad & Cost(x) = ay + bx \\ \text{s.t.} \quad & x \leq My \\ & x \geq 0, y \in \{0, 1\} \end{aligned} \qquad \text{(FC MILP)}$$

MODELING WITH MILPs V

4. DEPENDENT DECISIONS:

- Neither or both events/decisions 1 and 2 must occur. This is expressed as:

$$y_1 = y_2,$$

i.e., they are either both 0 or both 1.

- Decision/event 2 can only occur if event 1 occurs. This is expressed as:

$$y_2 \leq y_1.$$

i.e., y_2 can only be nonzero if $y_1 = 1$.

- Let x be a continuous activity that can be conducted at any level $0 \leq x \leq M$, where M is data. Suppose this activity can only be undertaken if an event y occurs. This is represented by:

$$x \leq My,$$

i.e., if $y = 0$ then $x = 0$, else if $y = 1$ then x can be chosen $0 \leq x \leq M$.

MODELING WITH MILPs VI

Facility-location problem:

(From Bertsimas, Tsitsiklis Ch. 10):

Suppose we are given n potential facility locations and a list of m clients who need to be serviced from these locations. There is a fixed costs c_j of opening a facility at location j, while there is a cost $d_{i,j}$ of serving client i from facility j. Goal: Select a set of facilities and assign each client to one facility, while minimizing total cost.

MODELING WITH MILPs VII

Logical Relationships: There is a standard polynomial time procedure (ref. Clocksin and Mellish 1981) for modeling boolean constraints and transforming these to MILP problems: (See Floudas MINLP Fundamentals book P. 237)

MODELING WITH MILPs VIII

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Example (Boolean Satisfiability)

Suppose we want to represent the fact: "*If either the adsorber unit or the membrane unit is selected in the design then do not use a cryogenic distillation unit in the design*." If this is represented with *Boolean* (logical) variables that can take the values TRUE or FALSE:

$$P_A = \text{the adsorber unit is selected in the design}$$
$$P_M = \text{the membrane unit is selected in the design}$$
$$P_{CD} = \text{the cryogenic distillation unit is selected in the design}$$

then the verbal statement can be expressed in classical propositional logic as:

$$(P_A \vee P_M) \Rightarrow \neg P_{CD}$$

Finding a set of logical values for P_A, P_M and P_{CD} that make this proposition true is an instance of the Boolean satisfiability problem.

MODELING WITH MILPs IX

There is a standard (polynomial time) procedure to convert any logical proposition into a set of linear inequalities in terms of binary variables.

1. Replace implication by its equivalent disjunction, i.e.:

$$(P_1 \Rightarrow P_2) \Leftrightarrow \neg P_1 \vee P_2.$$

   So, for the example

$$\neg(P_A \vee P_M) \vee \neg P_{CD}.$$

2. Apply DeMorgan's Theorem to put the negation inwards.

$$\neg(P_1 \wedge P_2) \Leftrightarrow \neg P_1 \vee \neg P_2$$

$$\neg(P_1 \vee P_2) \Leftrightarrow \neg P_1 \wedge \neg P_2.$$

   For the example:

$$(\neg P_A \wedge \neg P_M) \vee \neg P_{CD}.$$

MODELING WITH MILPs X

3. Distribute the logical operator OR over the logical operator AND operator recursively using the equivalence:

$$(P_1 \wedge P_2) \vee P_3 \Leftrightarrow (P_1 \vee P_3) \wedge (P_2 \vee P_3).$$

   For the example:

$$(\neg P_A \vee \neg P_{CD}) \wedge (\neg P_M \vee \neg P_{CD}).$$

   This yields the *Conjunctive Normal Form* of the proposition; a conjunction of disjunctions.

4. Introduce a binary variable corresponding to each Boolean variable: For the example:

$$y_A = \begin{cases} 1 & \text{if } P_A = \text{ TRUE} \\ 0 & \text{if } P_A = \text{ FALSE.} \end{cases}$$

   Note that $(1 - y_A)$ corresponds to $\neg P_A$.

MODELING WITH MILPs XI

5. Use the following standard transformations between elementary Boolean propositions to linear inequalities in terms of binary variables.

$$
\begin{aligned}
P_1 \vee P_2 \vee \ldots \vee P_n &\qquad y_1 + y_2 + \cdots + y_n \geq 1 \\
P_1 \wedge P_2 \wedge \ldots \wedge P_n &\qquad y_1 \geq 1,\ y_2 \geq 1, \ldots, y_n \geq 1 \\
P_1 \Rightarrow P_2 &\qquad y_2 \geq y_1 \\
P_1 \Leftrightarrow P_2 &\qquad y_2 = y_1 \\
P_1 \otimes P_2 \otimes \ldots \otimes P_n &\qquad y_1 + y_2 + \cdots + y_n = 1
\end{aligned}
$$

where $\otimes$ is the exclusive or operator. Using these identities any proposition in conjunctive normal form can be automatically converted to linear inequalities in terms of binary variables. Specifically, each disjunction in the conjunctive normal form must be independently true, yielding:

$$
\begin{aligned}
(1 - y_A) + (1 - y_{CD}) &\geq 1 \\
(1 - y_M) + (1 - y_{CD}) &\geq 1.
\end{aligned}
$$

These would then become constraints of an MILP problem.

LP relaxation of an MILP problem I

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Definition (LP Relaxation)

Consider the 0-1 MILP:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & \mathbf{c}^{\mathrm{T}}\mathbf{x} + \mathbf{d}^{\mathrm{T}}\mathbf{y} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}_{+}^{n_x},\ \mathbf{y} \in \{0,1\}^{n_y} \end{aligned} \tag{0-1 MILP}$$

its LP relaxation is:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & \mathbf{c}^{\mathrm{T}}\mathbf{x} + \mathbf{d}^{\mathrm{T}}\mathbf{y} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}_{+}^{n_x},\ \mathbf{y} \in [0,1]^{n_y}. \end{aligned} \tag{LP-R}$$

Exercise : Let z_{MILP},z_{LP-R} be corresponding global optimal values, what's their relationship? How about the objective value of LP-R at a feasible (x, y). Infeasible?

TIGHT MILP formulations I

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Key Points:

1. The feasible set of Problem LP-R is larger than the feasible set of Problem 0-1 MILP. From the **Lower Bounding Principle**, f^*_{LP-R} yields a LBD on f^*_{MILP}.
2. For faster convergence, we want the LBD from f^*_{LP-R} to be as close as possible to the true f^*_{MILP}, thus we want the LP relaxation to be as **tight** as possible i.e., the **Integrality gap** to be as small as possible.
3. So counter-intuitively, **more constraints for the MILP result in faster convergence** usually (because these yield tighter LP relaxations). It is desirable to add constraints! In fact, certain algorithms e.g., cutting planes are redundant valid constraints
4. Usually it is a good idea to calculate tight values for bounds and make the M in Big-M constraints as small as possible

Tight Formulations - Example I

Consider MILP-A:

$$
\begin{array}{llrcrcrcrcl}
 & & & \min\limits_{\mathbf{y}\in\mathbb{Z}^2_+} & y_1 & + & y_2 & & & & \\
\text{s.t.} & g_1(\mathbf{y}) & = & -y_1 & - & y_2 & + & 1 & \leq & 0 \\
 & g_2(\mathbf{y}) & = & y_1 & + & y_2 & - & 3 & \leq & 0 \\
 & g_3(\mathbf{y}) & = & -y_1 & + & y_2 & - & 1 & \leq & 0 \\
 & g_4(\mathbf{y}) & = & y_1 & - & y_2 & - & 1 & \leq & 0.
\end{array}
\qquad \text{(MILPA)}
$$

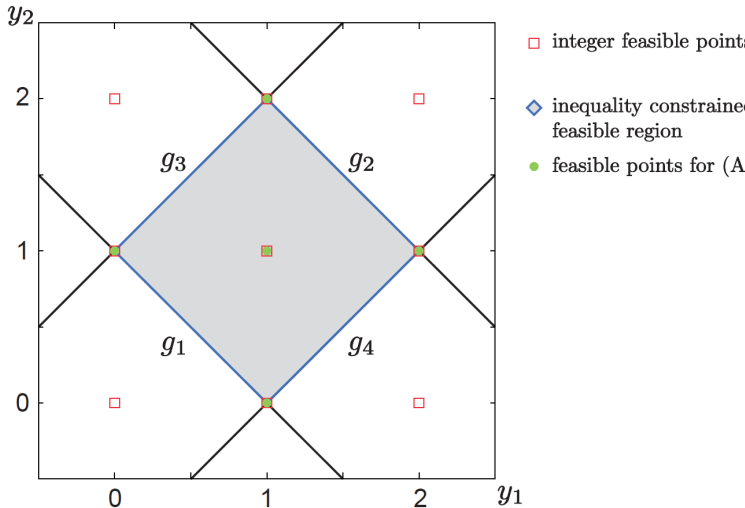

Figure 7.1: Feasible points for MILPA

Now consider MILP-B

$$
\begin{array}{llcrcrcrcl}
 & & & \min\limits_{\mathbf{y}\in\mathbb{Z}^2_+} & y_1 + y_2 & & & & & \\
\text{s.t.} & \hat{g}_1(\mathbf{y}) & = & -y_1 & - & y_2 & + & 0.5 & \leq & 0 \\
 & \hat{g}_2(\mathbf{y}) & = & y_1 & + & y_2 & - & 3.5 & \leq & 0 \\
 & \hat{g}_3(\mathbf{y}) & = & -y_1 & + & y_2 & - & 1.5 & \leq & 0 \\
 & \hat{g}_4(\mathbf{y}) & = & y_1 & - & y_2 & - & 1.5 & \leq & 0
\end{array}
\qquad \text{(MILPB)}
$$

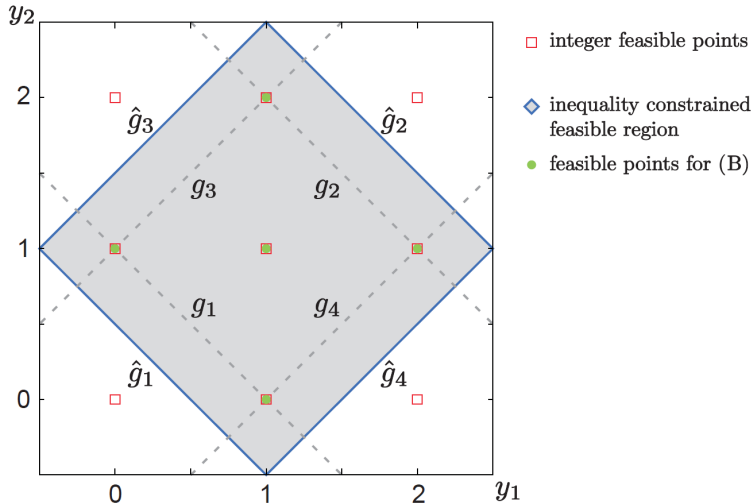

Figure 7.2: Feasible points for MILPB

Tight Formulations - Example I

- MILP-A and MILP-B are essentially the same problem
- By inspection, what is the optimum and the optimal value of the MILP?
- By inspection, what is the integrality gap of the corresponding LP relaxations?
- Which formulation is tighter and quicker to solve?

Branch-and-Bound Tree I

- Basically a divide an conquer approach.

Problem (P 7.9)

Solve the 0-1 MILP:

$$
\begin{aligned}
\min_{x,\mathbf{y}} \quad & x + y_1 + 3y_2 + 2y_3 \\
\text{s.t.} \quad & -x + 3y_1 + 2y_2 + y_3 \leq 0 \\
& -5y_1 - 8y_2 - 3y_3 \leq -9 \\
& x \geq 0, \mathbf{y} \in \{0,1\}^3
\end{aligned}
$$

with the branch-and-bound algorithm. Draw the branch-and-bound tree, label each node with the order in which it is explored, and show the action taken at each node and the lower bound computed.
For this question it is advised that you use software such as GAMS to solve the LP subproblems. However, do not use GAMS or other software for the branch-and-bound algorithm; the purpose of this question is for you to do this yourself. You can use GAMS to check that your implementation produces a correct answer.

Explicit Enumeration I

Implicit Enumeration I

Theorem (Feasibility Principle - Prop. 2.11 Barton Book P.20)

Let $D, E \subset \mathbb{R}^n$ and suppose that $D \subset E$.

1. *If $\min_{\mathbf{x} \in E} f(\mathbf{x})$ is infeasible then $\min_{\mathbf{x} \in D} f(\mathbf{x})$ is infeasible.*
2. *Otherwise, if $\mathbf{x}^*$ is an optimal solution for $\min_{\mathbf{x} \in E} f(\mathbf{x})$ and $\mathbf{x}^* \in D$ then $\mathbf{x}^*$ is also an optimal solution for $\min_{\mathbf{x} \in D} f(\mathbf{x})$.*

Note: Feasibility set of each descendant node is a subset of the feasibility set of current node.

Qn: How would you expect the optimal values of the nodes to behave as we go down the tree? Why? Hint: LBP

Implicit Enumeration II

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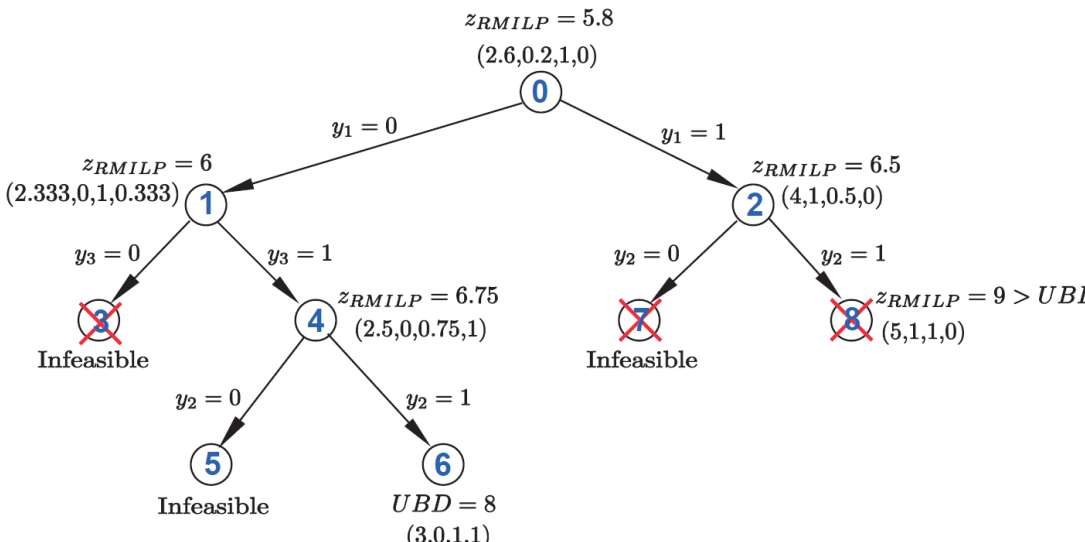
Fathoming/Pruning strategies

1. Infeasibility - If a node is infeasible, all descendants are infeasible.
2. Optimality - If the optimal solution of a node if integer feasible (i.e., it is feasible in the descendant nodes), then it is also optimal in the descendant nodes.
3. Value Dominance (the most important) - From the **Lower Bounding Principle**, you would only expect the optimal values of the descendants to get larger (since we are always restricting). Thus, if we have an incumbent UBD value for the overall MILP is lower than the optimal value of a node, we cannot hope to find a smaller UBD (i.e., improve this UBD) from any descendants. Since the descendants don't give any useful information, we can prune them all.

Keep track:
For the overall MILP, what gives LBDs: z_{LP-R} optimal value at root node
UBDs: z_{node} if optimal value is integer feasible (i.e., when we prune by optimality)

Answer: Branch-and-Bound tree I

Answer: Branch-and-Bound tree II